

Project Title: Snapdragon Edge-Compute Ground Station for Autonomous UAVs

The Problem: Autonomous drones and UAVs operate under strict payload and battery limitations. Carrying heavy onboard processors for advanced AI vision tasks (like obstacle avoidance) drastically reduces flight time. Alternatively, streaming live video to cloud servers introduces dangerous latency, consumes high bandwidth, and fails completely in remote field operations without internet access.

The Solution: We transform a Snapdragon-powered HP PC into a powerful, localized "AI Brain" on the ground. Instead of the drone doing the heavy computational lifting, it transmits a lightweight live camera feed to the laptop via a telemetry radio. The PC processes this feed entirely offline using its NPU. Once a decision is made, the laptop instantly transmits lightweight flight commands (e.g., "EMERGENCY STOP") back to the drone's flight controller.

Technical Implementation:

- **Qualcomm AI Hub:** We utilize a pre-quantized computer vision model (YOLOv8/11) downloaded directly from the Qualcomm AI Hub.
- **Hardware Acceleration:** The model runs natively on the Snapdragon Hexagon NPU using ONNX Runtime. This ensures maximum inference speed while keeping the CPU and GPU entirely free for other ground station tasks.
- **The Bridge:** A Python script processes the NPU's bounding box outputs and translates them into physical serial commands sent to a connected radio transmitter (e.g., nRF24L01).

Impact & Innovation: By offloading AI vision to the Snapdragon NPU, the drone remains lightweight and power-efficient. The system operates with zero cloud connectivity, ensuring absolute data privacy and ultra-low latency. This project perfectly demonstrates how Windows on ARM can be utilized as a robust edge-computing hub for real-world robotics and hardware integration.